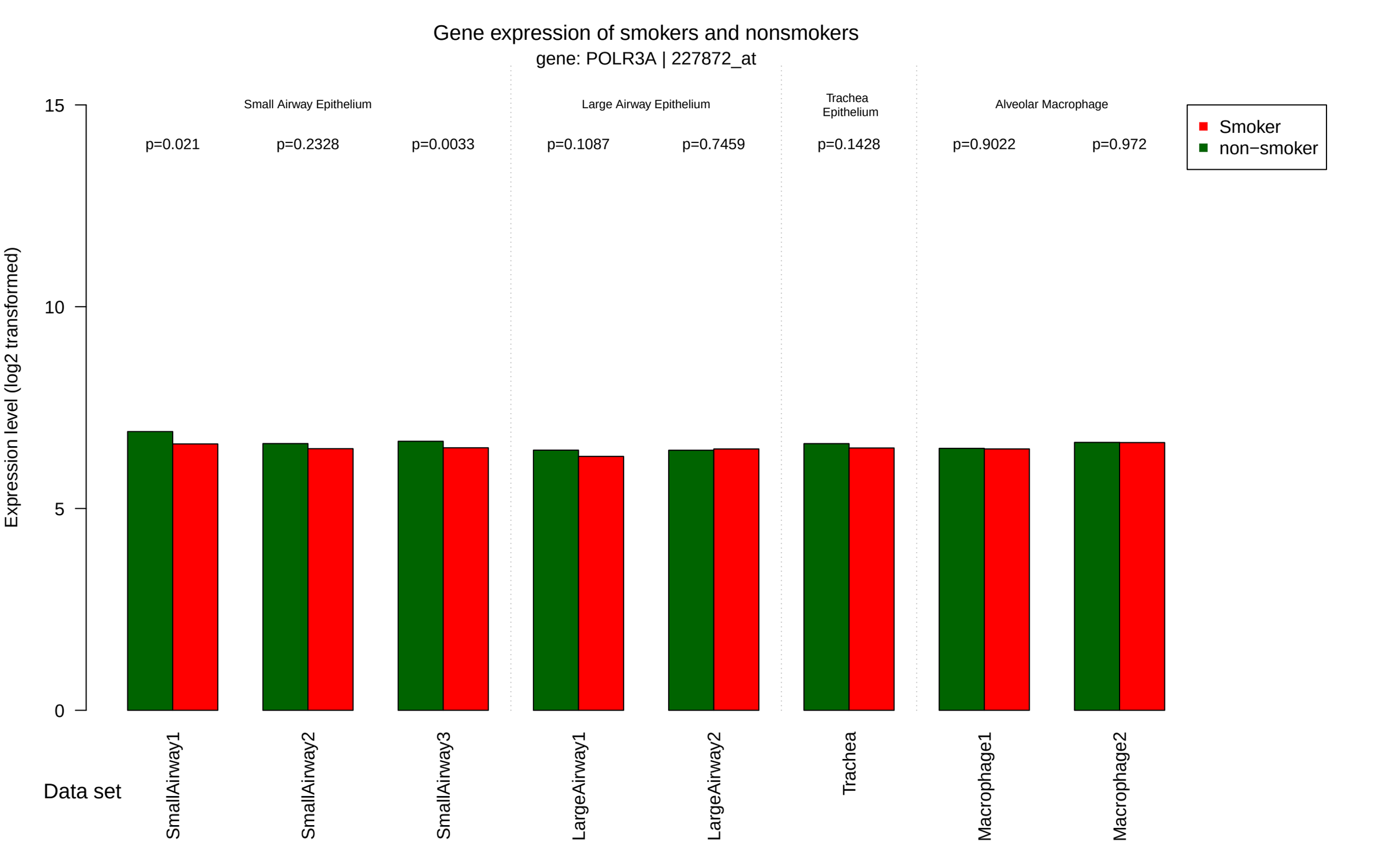
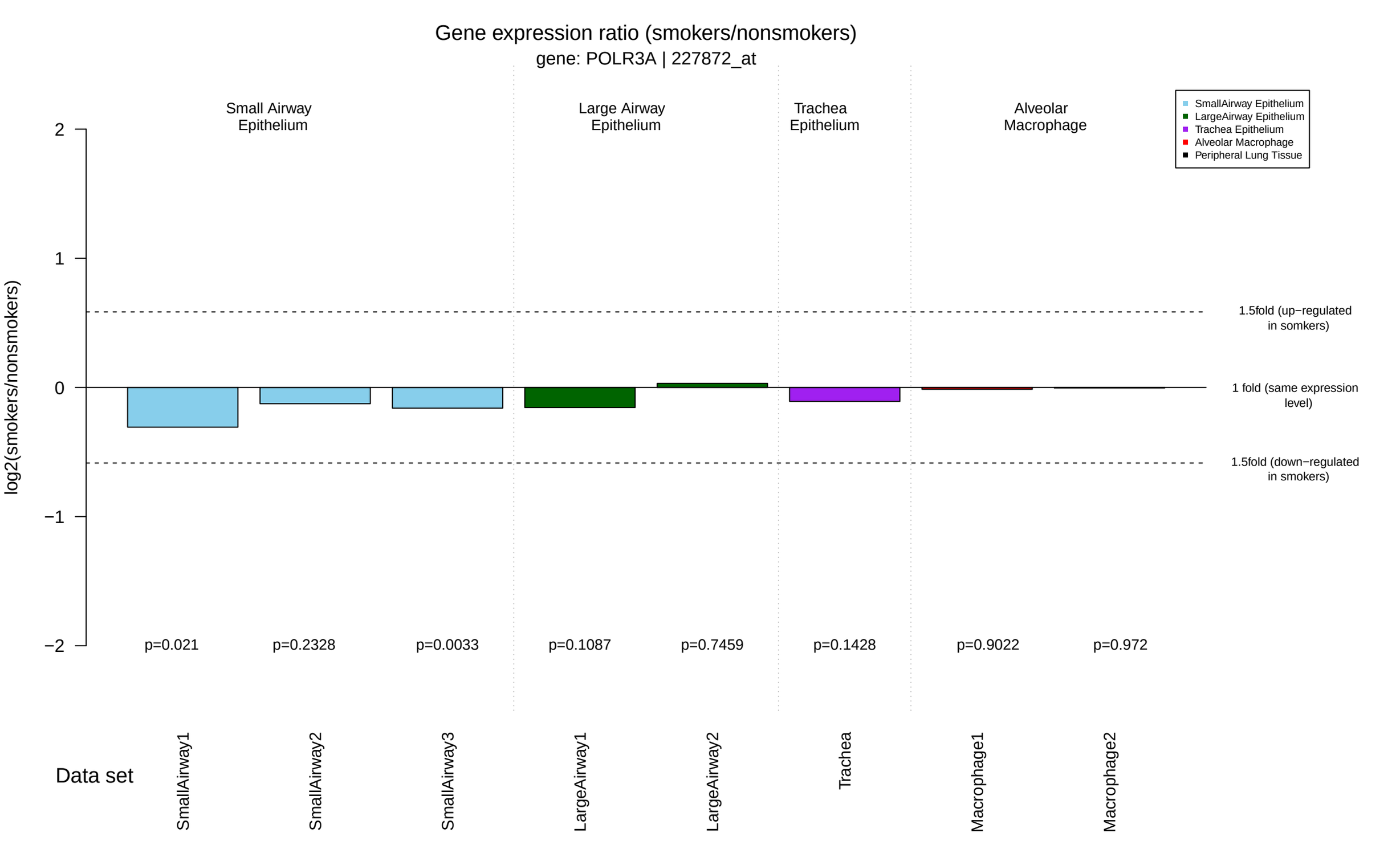
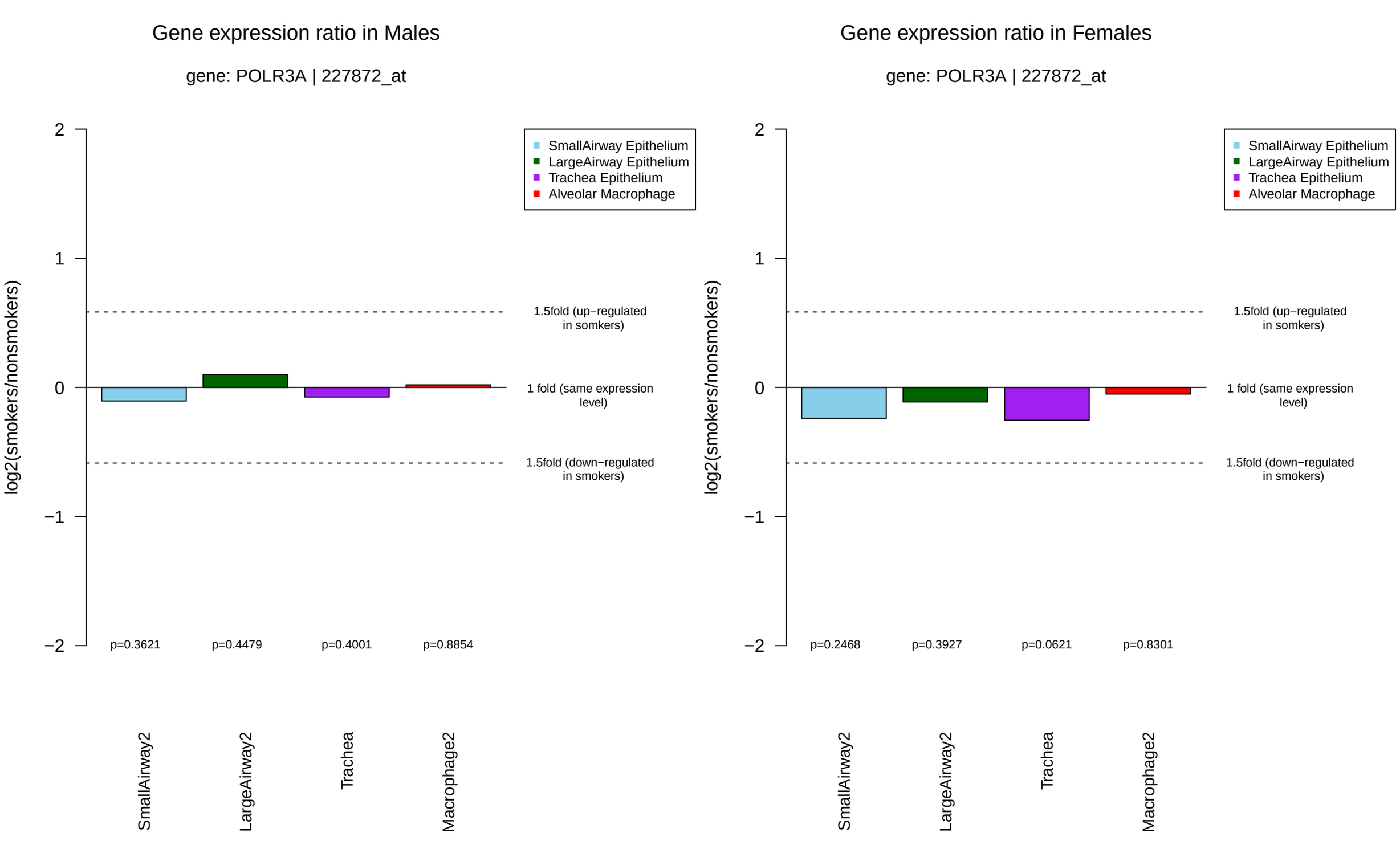
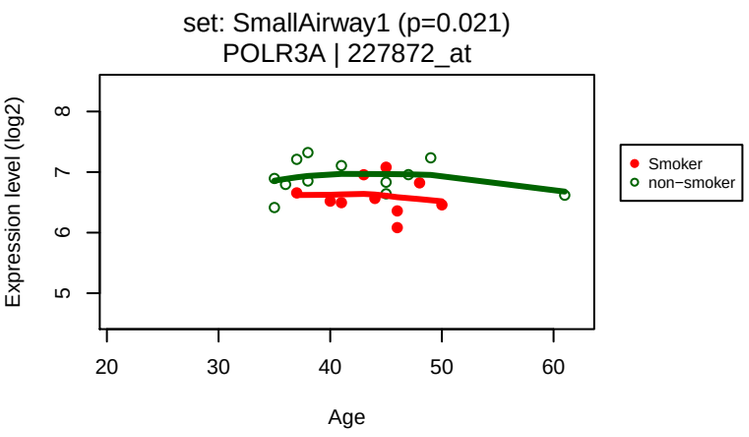
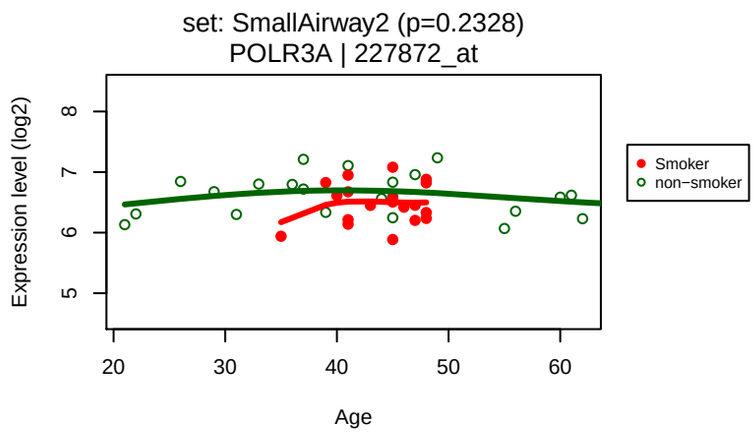
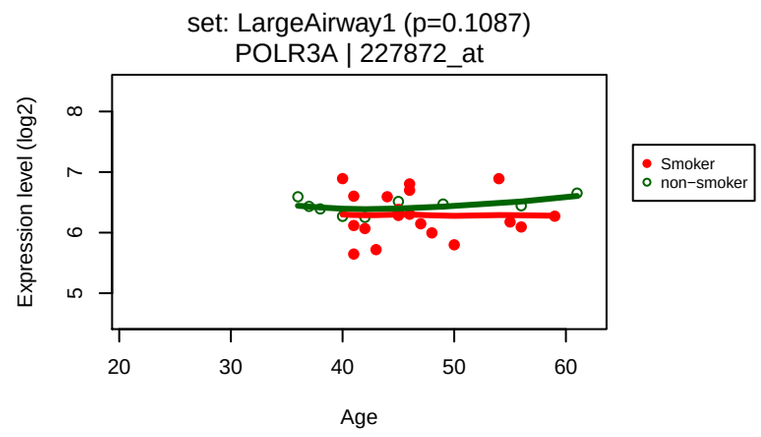

Gene expression of smokers and nonsmokers
gene: POLR3A | 227872_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.021
p=0.2328
p=0.0033
p=0.1087
p=0.7459
p=0.1428
p=0.9022
p=0.972
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

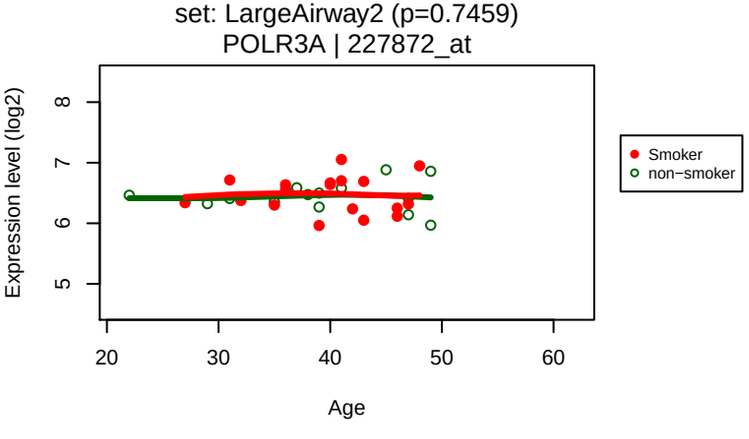

Gene expression ratio (smokers/nonsmokers)
gene: POLR3A | 227872_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.021
p=0.2328
p=0.0033
p=0.1087
p=0.7459
p=0.1428
p=0.9022
p=0.972
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

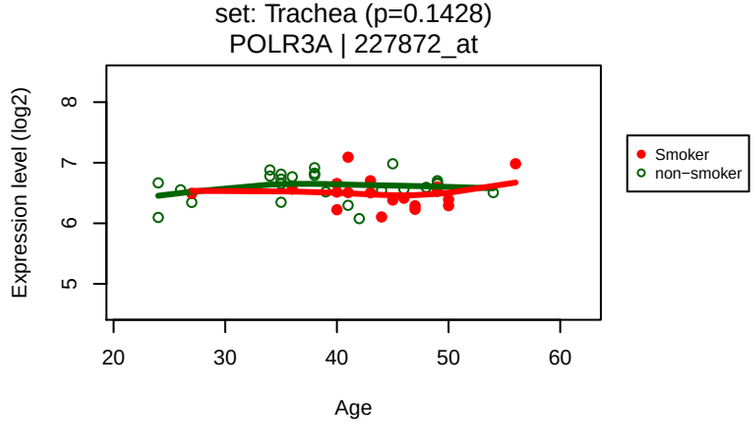

Gene expression ratio in Males
gene: POLR3A | 227872_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3621
p=0.4479
p=0.4001
p=0.8854
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR3A | 227872_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2468
p=0.3927
p=0.0621
p=0.8301
SmallAirway2
LargeAirway2
Trachea
Macrophage2

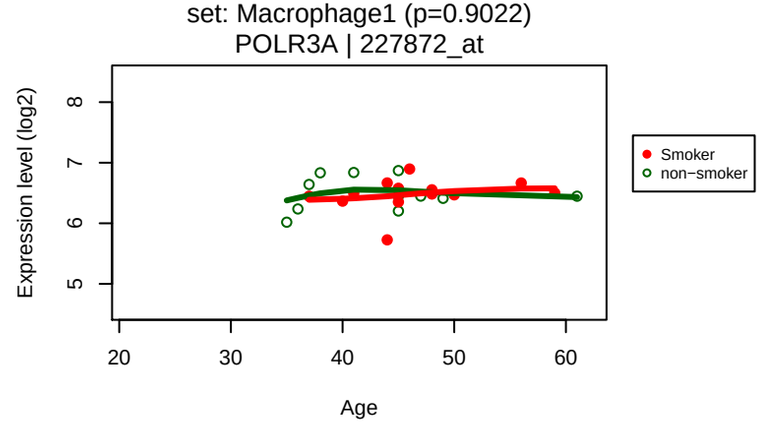

set: SmallAirway1 (p=0.021)
POLR3A | 227872_at
Expression level (log2)
Age
Smoker
non−smoker

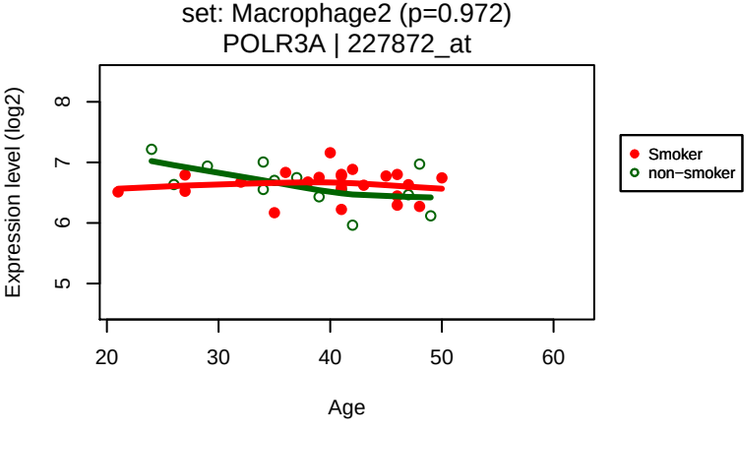

set: SmallAirway2 (p=0.2328)
POLR3A | 227872_at
Expression level (log2)
Age
Smoker
non−smoker

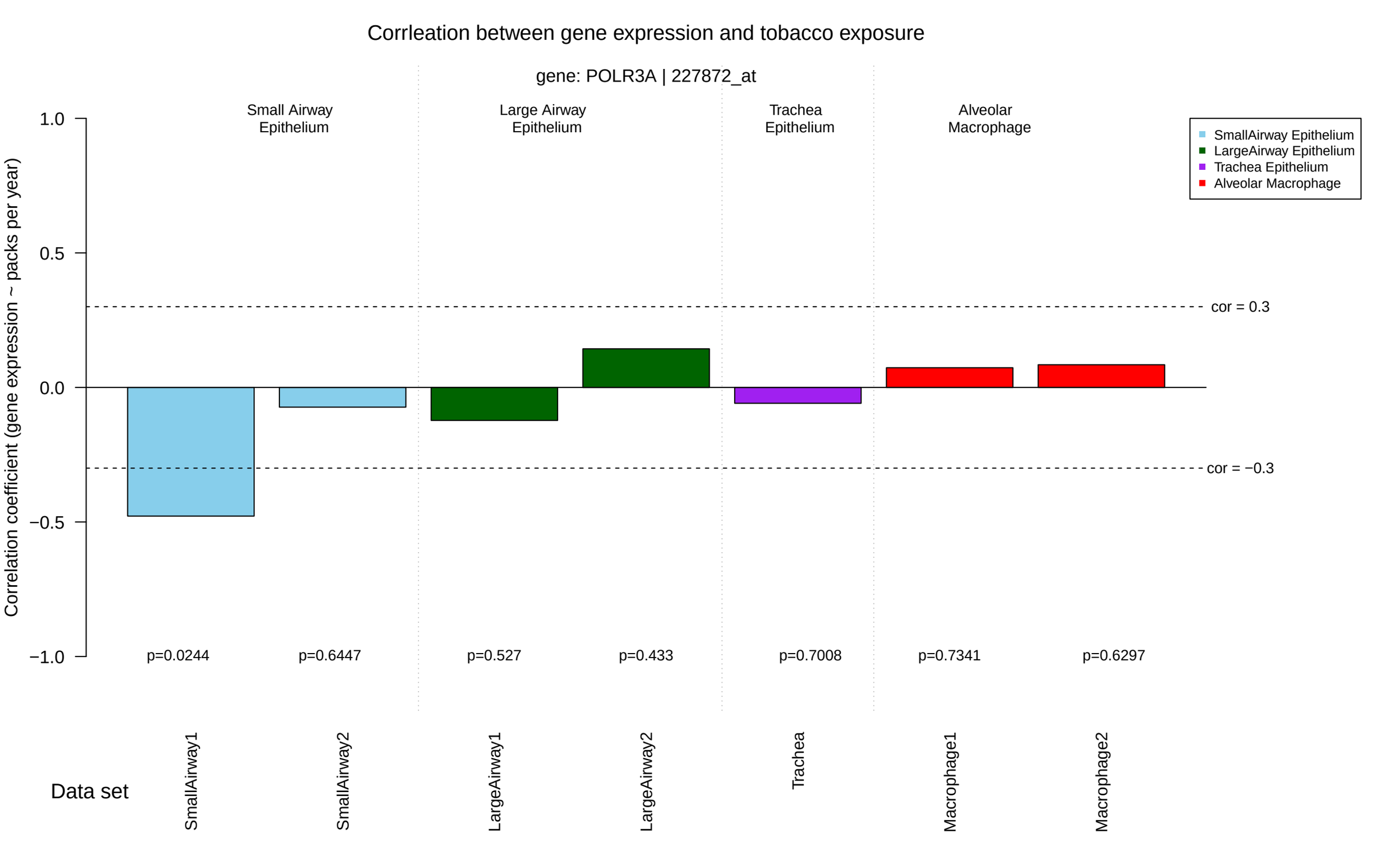

set: LargeAirway1 (p=0.1087)
POLR3A | 227872_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7459)
POLR3A | 227872_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1428)
POLR3A | 227872_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9022)
POLR3A | 227872_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.972)
POLR3A | 227872_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR3A | 227872_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0244
p=0.6447
p=0.527
p=0.433
p=0.7008
p=0.7341
p=0.6297
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2